OneWelbeck ENT

OneWelbeck ENT, 1 Welbeck Street, London, W1G 0AR
Tel: 020 3653 2007

03/04/2022

Dear Doctor,

Re: Ms Jane Smith, Date of Birth: 15/03/1975, Patient ID: 1000001

Thank you for referring Ms Jane Smith for a vestibular assessment. The following report summarises the findings from her evaluation conducted on 03/04/2022 at OneWelbeck ENT.

A comprehensive vestibular test battery was performed including cervical vestibular evoked myogenic potentials (cVEMP), video head impulse testing (vHIT), computerised dynamic posturography (mCTSIB), and videonystagmography (VNG) including caloric irrigation.

cVEMP: Cervical VEMPs were recordable bilaterally. Latencies and amplitudes were within normal limits for age. Asymmetry ratio was within acceptable range.

Video Head Impulse Test (vHIT): Horizontal semicircular canal gain was within normal limits bilaterally. No significant corrective saccades were identified on head impulse in the yaw plane.

mCTSIB (Balance): Performance on modified clinical test of sensory integration and balance was within normal limits across all four conditions.

VNG: Smooth pursuit gain was mildly reduced bilaterally, which may be consistent with central pathology or medication effect. Optokinetic responses were symmetrical. No spontaneous or gaze-evoked nystagmus was noted. Caloric responses were present bilaterally with a canal paresis of 12% (within normal limits < 25%).

Overall, the vestibular assessment did not reveal significant peripheral vestibular dysfunction. The mild reduction in smooth pursuit gain warrants monitoring.

Please do not hesitate to contact us should you have any queries regarding these results.

Patient: **Ms Jane Smith** DOB: **15/03/1975** ID: **1000001**

Recommendations:

1. Given the finding of mildly reduced smooth pursuit gain, a neurological opinion may be considered if clinically indicated.

2. The patient may benefit from a course of vestibular rehabilitation exercises to address any residual instability complaints.

3. Repeat vestibular assessment in 12 months or sooner if symptoms progress.

Yours sincerely,

Dr A. Kumar
Audiologist / Vestibular Scientist
OneWelbeck ENT
OneWelbeck ENT, 1 Welbeck Street, London, W1G 0AR

Cervical Vestibular Evoked Myogenic Potentials (cVEMP)

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
Age	47 yrs	**Date**	03/04/2022	**Audiologist**	Dr A. Kumar

cVEMP Waveforms — Right and Left

[cVEMP bilateral waveform traces — Right (red) / Left (blue)]

Parameter	Right	Left	Normal Range
P1 Latency (ms)	13.2	13.5	10–18 ms
N1 Latency (ms)	23.4	23.8	18–28 ms
P1–N1 Amplitude (µV)	142	138	> 50 µV
Asymmetry Ratio (%)	1.4	—	< 33%
Response	Present	Present	—

Interpretation: cVEMP responses present bilaterally. Latencies and amplitudes within normal limits. Asymmetry ratio normal. No significant saccular dysfunction detected.

cVEMP — EMG Recordings & Settings

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
Age	47 yrs	**Date**	03/04/2022	**Audiologist**	Dr A. Kumar

EMG Envelope Curves

[EMG rectified curves — tonic contraction levels]

Setting	Value
Stimulus type	Tone burst 500 Hz
Stimulus level	95 dB nHL
Filter band-pass	10–1500 Hz
Averaging sweeps	200
Electrode montage	SCM ipsilateral
Amplifier gain	5000×

Vestibular Assessment — Video Head Impulse Test (vHIT)

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Horizontal Canal VOR Gain (Yaw Plane)

[Head impulse velocity traces — Right (R) and Left (L) canals]

Canal	VOR Gain	Normal Range	Corrective Saccades
Right Horizontal	0.95	0.80–1.20	None
Left Horizontal	0.92	0.80–1.20	None

Interpretation: VOR gain within normal limits bilaterally in the horizontal (yaw) plane. No overt or covert corrective saccades identified. No significant horizontal canal dysfunction detected.

mCTSIB — Computerised Balance Assessment

Patient: Ms Jane Smith	DOB: 15/03/1975	ID: 1000001	Age: 47 yrs	Date: 03/04/2022

Height	5'4"	**Weight**	138.0 lbs	**BMI**	23.6

Balance Trial Results — Centre of Gravity Sway

[mCTSIB sway area charts — 4 conditions]

Condition	Sway Area (cm^2)	Sway Velocity (cm/s)	Result
Firm surface / Eyes open	1.8	0.9	Normal
Firm surface / Eyes closed	3.2	1.4	Normal
Foam surface / Eyes open	4.1	1.8	Normal
Foam surface / Eyes closed	9.8	3.2	Normal

Interpretation: Balance performance within normal limits across all four mCTSIB conditions.

Videonystagmography (VNG) — Summary and Clinical Comments

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
Age	47 yrs	**Date**	03/04/2022	**Audiologist**	Dr A. Kumar

Clinical Notes / Examiner Comments:

Patient: Ms Jane Smith DOB: 15/03/1975 Exam date: 03/04/2022

Patient is a 47-year-old female referred for dizziness and imbalance of 6 months duration. No significant past medical history. Not on ototoxic medication.

VNG performed using Interacoustics EyeSeeCam system. Patient tolerated all subtests without difficulty. No spontaneous nystagmus noted in primary gaze with fixation removed. No gaze-evoked nystagmus. Smooth pursuit mildly asymmetric — gain reduced L > R (see smooth pursuit report). Optokinetic responses symmetric. Caloric irrigation performed with warm (44°C) and cool (30°C) water. Canal paresis 12% (Right > Left, within normal limits). Directional preponderance 8% (within normal limits).

Examiner: ADM / Administrator

VNG Test Report — Subtest Overview

Patient	Ms Jane Smith	DOB	15/03/1975	Patient ID	1000001
Age	47 yrs	Date	03/04/2022	Audiologist	Dr A. Kumar

Subtest	Result	Laterality	Notes
Calibration	Pass	Bilateral	Gain within normal limits
Gaze (Center)	Normal	—	No gaze nystagmus
Gaze (Left 20°)	Normal	—	No gaze nystagmus
Gaze (Right 20°)	Normal	—	No gaze nystagmus
Smooth Pursuit	Mildly reduced	Bilateral (L > R)	Gain 0.72 L / 0.78 R
Random Saccade	Normal	Bilateral	Latency and velocity within limits
Optokinetic	Normal	Symmetric	No asymmetry
Caloric (44°C R)	Present	Right	SPV 22°/s
Caloric (44°C L)	Present	Left	SPV 18°/s
Caloric (30°C R)	Present	Right	SPV 20°/s
Caloric (30°C L)	Present	Left	SPV 16°/s
Canal Paresis	12%	—	Normal (< 25%)
Directional Preponderance	8%	—	Normal (< 30%)

VNG — Calibration

Patient	Ms Jane Smith	DOB	15/03/1975	Patient ID	1000001
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Eye Position Calibration Charts

[Calibration H — Horizontal eye position trace]

[Calibration V — Vertical eye position trace]

[Calibration gain plot — H]

[Calibration gain plot — V]

VNG — Gaze Testing

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
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Center Gaze

[Center Gaze — eye position trace (no nystagmus)]

Left Gaze 20°

[Left Gaze 20° — eye position trace (no nystagmus)]

Right Gaze 20°

[Right Gaze 20° — eye position trace (no nystagmus)]

VNG — Smooth Pursuit

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
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[Smooth pursuit eye-target position overlay — all frequencies]

Frequency (Hz)	Gain Right	Gain Left	Symmetry (%)
0.2	0.85	0.82	98
0.4	0.80	0.74	92
0.6	0.78	0.72	89
Average	0.81	0.76	93

Interpretation: Smooth pursuit gain mildly reduced bilaterally (L slightly more than R). May reflect a non-specific central finding or medication effect. No gross asymmetry.

VNG — Random Saccade

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
Age	47 yrs	**Date**	03/04/2022	**Audiologist**	Dr A. Kumar

[Random saccade eye position and velocity traces]

Parameter	Right	Left	Normal
Latency (ms)	185	192	< 250 ms
Peak Velocity (°/s)	410	398	200–600 °/s
Accuracy (%)	96	94	80–120%

Interpretation: Saccade latency, peak velocity and accuracy all within normal limits.

VNG — Optokinetic Nystagmus (OKN)

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
Age	47 yrs	**Date**	03/04/2022	**Audiologist**	Dr A. Kumar

[OKN gain symmetry chart — Rightward vs Leftward stimulus]

Direction	Gain	SPV (°/s)	Result
Rightward	0.82	33	Normal
Leftward	0.80	32	Normal
Asymmetry	1.2%	—	Normal (< 20%)

Interpretation: Optokinetic responses symmetric and within normal limits.

VNG — Caloric Testing

Patient	Ms Jane Smith	**DOB**	15/03/1975	**Patient ID**	1000001
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Caloric Irrigation — Pod View

[Caloric pod summary — 4 irrigations (44R, 44L, 30R, 30L)]

Caloric Response Bars (Maximum SPV)

[Bar chart — max SPV per irrigation condition]

Irrigation	Max SPV (°/s)	Peak Time (s)	Duration (s)
44°C Right (Warm R)	22	75	130
44°C Left (Warm L)	18	78	125
30°C Right (Cool R)	20	80	128
30°C Left (Cool L)	16	82	120

Derived Measure	Value	Normal Limit	Interpretation
Canal Paresis (Jongkees)	12%	< 25%	Normal — no significant unilateral weakness
Directional Preponderance	8%	< 30%	Normal — no significant preponderance
Total Caloric Response (R)	42°/s	> 12°/s	Normal
Total Caloric Response (L)	34°/s	> 12°/s	Normal

Interpretation: Caloric responses present bilaterally. No significant canal paresis or directional preponderance. No peripheral caloric asymmetry detected.

End of vestibular assessment report — Ms Jane Smith (15/03/1975) — assessed 03/04/2022 — Dr A. Kumar, OneWelbeck ENT